

Narrative-Integrated Thematic Analysis (NITA): AI-Supported Theme Generation Without Coding

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Abstract. The coding approach has dominated qualitative data analysis (QDA) since at least the introduction of grounded theory in the 1960s. The development of generative AI (GenAI) has led to an emerging debate about whether GenAI can help qualitative researchers shift away from the coding paradigm. Yet, the application of GenAI in QDA to date has largely focused on replacing human coding to improve efficiency and effectiveness, while the attention to facilitating the transition to alternative methods remains limited. In this article, we introduce a Narrative-Integrated Thematic Analysis procedure (NITA) that allows qualitative researchers to design, train and guide a GenAI in conducting thematic analysis. This NITA approach is regarded as a thematic analysis approach that positions researchers' reflexivity, intellect and judgement at the centre of the analysis process. This approach combines a reflexive, iterative monitoring, evaluation and learning procedure (PERFECT) with a conversational method for interacting with and guiding GenAI. Using an publicly available interview dataset provided by Lumivero and custom ChatGPT, we applied this approach through six stages: planning an initial PERFECT procedure, preparation, generating

candidate themes, constructing individual narratives, constructing meta-narrative, and writing up. Compared to our previous experiment on the same dataset using the coding approach, we have found that the themes generated this time had similar qualities. This reveals the transformative potential of GenAI to redefine thematic analysis.

Keywords: ChatGPT, Coding, Categorizing, Generative AI, Qualitative Data Analysis, Thematic Analysis, Template Analysis, Reflexivity

1 Introduction

“Any researcher who wishes to become proficient at doing qualitative analysis must learn to code well and easily. The excellence of the research rests in large part on the excellence of the coding” (Strauss, 1987, p.27).

Coding is a crucial method that refers to assigning a label that ‘symbolically assigns a summative, salient, essence-capturing, and/or evocative attribute for a portion of language based or visual data’ (Saldaña, 2016, p.4). Coding helps researchers deal with a vast amount of data and condense them into a manageable scale from which researchers can construct meaningful patterns (categories or themes) or theories (Miles & Huberman, 1994; Miles et al., 2013).

The statement above by Anselm Strauss, one of the creators of grounded theory, indicates the importance of coding to qualitative data analysis (QDA). Since its first publications, grounded theory has advocated for an extensive coding process which includes different coding methods such as open/ initial coding, axial coding, focused coding, and theoretical coding. The popularity of grounded theory and its coding methods (Glaser &

Strauss, 1967; Strauss & Corbin, 1991; Charmaz, 2006) has led students and newcomers in qualitative research to mistakenly believe that coding is a must in QDA process. The coding approach has dominated the field of QDA since 1980s with many data analysis approaches incorporating coding as a core component immediately after familiarizing with the data. For instance, Interpretative Phenomenological Analysis (Smith, 2009) has conceptual coding as part of initial noting (Step 2) that assists researchers in interacting with data at a conceptual or more interpretative level. Template Analysis (King et al., 2018) has preliminary coding (Step 2), which centers on techniques such as hierarchical and parallel coding to structure data into initial templates. Framework Analysis (Gale et al., 2013) also consider coding (descriptive or conceptual) (Step 3) as a foundation to form an analytical or thematic framework (Step 4). Similarly, semantic and latent coding (Stage) are crucial in helping generate candidate themes (Stage 3) in Reflexive Thematic Analysis process (Braun & Clarke, 2022).

Despite its popularity, “coding is just *one* way of analyzing qualitative data, not *the* way” (Saldaña, 2013, p.2, original italics by Saldaña). In their theorizing of QDA, Maxwell and Chmiel (2014) consider the foundation of coding paradigm, namely, similarity-based relations, as just *one* strategy of QDA alongside contiguity. While the similarity-based strategy aims to capture “resemblances or common features; their identification is based on comparison, which can be independent of time and place”, the contiguity-based strategy focuses on “juxtaposition in time and space, the influence of one thing on another, or relations among parts of a text; their identification involves seeing actual connections between things, rather than similarities and differences” (Maxwell & Chmiel, 2014, p. 22). Representing the former, coding alongside thematic generation seeks to facilitate the condensation and abstraction process that produces codes, metacodes or categories, themes,

and at the highest level, theories, while narrative or case analyses seek to maintain the actual connections and contextual information provided by participants (Saldaña, 2016; Miles et al., 2013, Maxwell & Chmiel, 2014). Similarly, Freeman (2016), conceptualizes coding as part of a categorizing thinking mode alongside four other modes of thinking in qualitative data analysis including narrative thinking, dialectical thinking, poetical thinking, diagrammatical thinking. Apart from coding, researchers can use story construction, creating an aesthetic space to engage with data, exploration of contradictions and negotiations, or working with moving assemblages (or diagrams).

Thematic analysis (TA) refers to a family of methods, rather than a single, unified method (Terry et al., 2017). Braun and Clarke (2022) classifies TA in three main groups, running on a continuum from small q to Big Q poles (see Kidder & Fine (1987) for the original classification of small q/Big Q): coding reliability TA (Boyatzis, 1998; Fereday & Muir-Cochrane, 2006; Guest, MacQueen, & Namey, 2012), codebook TA including multiple forms of in-between small q and Big Q such as Template Analysis (King et al., 2018; Brooks & King, 2014), Matrix analysis (Miles & Huberman, 1994, Miles et al., 2013), Two-Cycle analysis (Saldaña 2013, 2016); Framework (Method) Analysis (Ritchie & Spencer, 1994; Gale et al. 2013), and the Reflexive Thematic Analysis (RTA) by Braun and Clarke (2006, 2022). Coding reliability TA is underlined by (post)positivism, RTA by constructivism, critical realism and relativism, codebook TA by realism, critical realism, constructionism, and pragmatism (Braun & Clarke, 2022; Ritchie & Spencer, 1994; King et al., 2018). While these approaches are different in their underlining epistemological and methodological assumptions, they all place coding at the foundation of their analysis processes.

Qualitative researchers have long used technologies, particularly, Computer-Assisted Qualitative Data Analysis Software (CAQDAS) in assisting their coding since 1980s (Nguyen-Trung, 2024a). Current CAQDAS, notably DiscoverText, QDA Miner, and Leximancer, have taken the advantage of the technological advancements of Artificial Intelligence (AI) such as Machine Learning (ML) in the last two decades to significantly assist human researchers in coding and classification of text data (Paulus, 2023; Shulman 2011; Silver 2019). Yet, the emergence of generative AI (GenAI) technology, particularly ChatGPT in late 2022, seems presenting a new turning point. There has been an upsurge in the application of GenAI in various qualitative analysis approaches. Thematic Analysis has received the most applications, particularly RTA (Morgan, 2023; Hitch, 2023; Hamilton et al., 2023, Drápal et al., 2023; De Paoli 2023, 2024; Prescott et al., 2024), and Template Analysis (Nguyen-Trung, 2024a). Additionally, there has been trials in other different approaches such as context analysis (Chew et al., 2023), grounded theory (Sinha et al., 2024; Wachinger et al., 2024). Despite differences, most of these applications more or less involved with transforming coding (i.e., assigning labels to data segments) in order to seek patterns in data (themes or categories). Different being the application that seeks to employ different discourse analysis techniques such as topic segmentation and dialogue discourse parsing (Fan et al., 2023), concordance analysis and function-to-form analysis (Curry, Baker, and Brookes, 2024).

Different from them, Morgan (2025) advocates for a non-coding method labelled ‘query-based approach’ (QBA). QBA focuses on asking GenAI to search for themes without coding. It includes three steps: Step 1: asking broad, unstructured queries; Step 2: following up with more specific queries, and Step 3: examining the supporting data (Morgan, 2025). In

comparing with other TA approaches, Morgan's approach concentrates on theme development and refinement (Steps 4 and 5 in Reflexive Thematic Analysis (RTA) by Braun and Clarke 2022), applying analytical framework in Framework Analysis (Gale et al. 2013), or modifying and defining template in Template Analysis (King et al., 2018) (see Fig. 5 for more information). In his paper, Hayes advocates for a similar approach that considers large language models (LLMs) as a 'conversational partner' through whom researchers can converse with qualitative data to identify "recurring patterns, subtle shifts in tone, or suggest new interpretive angles" including visualised images or diagrams (Hayes, 2025, p. 1). Reflecting on these recent developments, Friese (2025) suggests that with new AI technologies, qualitative research is having an opportunity about to take a revolutionary shift that can transform the way qualitative data is analyzed.

In this paper, we aim to innovate TA by advocating for the non-coding approach as pioneered by Morgan, Hayes and Friese. We aim to answer the key research question: Can ChatGPT/GenAI help generate themes without coding? The second question is to follow up with our previous work (Nguyen-Trung, 2024a) in which we used a coding-based methods - the Guided AI Thematic Analysis (GAITA) approach – to code with AI and generate themes using the dataset from the study "Environmental Change in Down East" (Lumivero, 2020; Campbell and Meletis 2011). In the current study, we want to see if our new non-coding approach, NITA, can generate similar themes as produced by the GAITA approach.

In the sections that follow, we start with describing our dataset and methodological approach. In Section 3, together with the analysis activities and results within each stage we also discuss these results with the existing TA approaches.

2 Narrative-Integrated Thematic Analysis (NITA) Approach

2.1 Context and dataset

In our previous work (Nguyen-Trung, 2024a), we presented an adapted version of Template Analysis (King et al., 2018) called Guided AI Thematic Analysis (GAITA). GAITA, developed based on pragmatism and the human in the loop approach, was used to guide GPT-4 in four stages of template analysis: data familiarization, preliminary coding, template formation and finalization, and theme development. In this experiment, we used the sample public project "Environmental Change in Down East" provided by Lumivero (2020). This sample project includes 11 short interviews conducted in 2008-2009 by scholars from Duke University's Nicholas School of the Environment at their Marine Lab in Beaufort, N.C, which aimed to examine how communities in the Down East region of Carteret County, North Carolina, USA, perceived the change in their environment (Campbell and Meletis 2011). Each interview transcript is about 6 pages long or about 2,000 words. Each transcript is structured by topics or headings including Q.1. Connection to Down East; Q.2. Connection to Down East natural environment; Q.3. Professional perspective on Down East; Q.4. Community and Environmental Change; Q.5. Vision for the future of Down East; and Q.6. Barriers to and opportunities for achieving your vision.

In the current study, to facilitate the comparison with our previous work (Nguyen-Trung, 2024a), we maintain the same research question "How do these people connect to Down East?" and utilise the same dataset. While our previous work (Nguyen-Trung, 2024a) employed GPT-4 to proceed Section Q1 from 11 interviews that made up a total of 6,306 words, in the current study, we include all 11 interviews (20,832 words) into our experiment

with GPT-4o. In addition, in the current experiment, we used custom GPT – a feature in ChatGPT that allows users to create and fine-tune their AI chatbots.

2.2 NITA - an approach to non-coding with GenAI

We develop Narrative-Integrated Thematic Analysis (NITA) as one method to facilitate non-coding approach to thematic analysis. This approach is considered as part of the thematic analysis (TA) family, but is specifically designed to engage with GenAI in generating themes without coding. Our NITA approach runs through six circular, iterative stages as seen in Fig 1.

- *Stage 1 - Planning the initial perfect procedure:* Purposefully plan the initial PERFECT (developing purpose, envisioning stages/steps to meet purposes, realizing GenAI-generated outputs, formulating prompts, experimenting with GenAI, checking results, and tuning/adapting) procedure to guide GenAI-supported qualitative analysis, linking research questions with AI tasks and expected outputs.
- *Stage 2 - Preparing data and the AI system:* Familiarize with the data, prepare the dataset, and choose GenAI tool and configure a customized GenAI system, including training the AI model to support thematic analysis effectively.
- *Stage 3 - Generating candidate themes:* Generate lists of candidate themes, discuss, and use one unified list for guiding analysis.
- *Stage 4 - Constructing individual narratives:* Apply the candidate themes to interpret and construct personal stories for each participant, maintaining real-world relationships (contiguity) and continuously refining themes through reflexive iteration.

- *Stage 5 - Constructing meta narratives:* Apply the refined themes construct patterns across groups (e.g., age, gender) to form meta-narratives, highlighting both shared experiences and divergent perspectives.
- *Stage 6 - Writing up:* Synthesize the constructed narratives, integrating thematic insights to address the research questions through coherent, reflexive reporting.

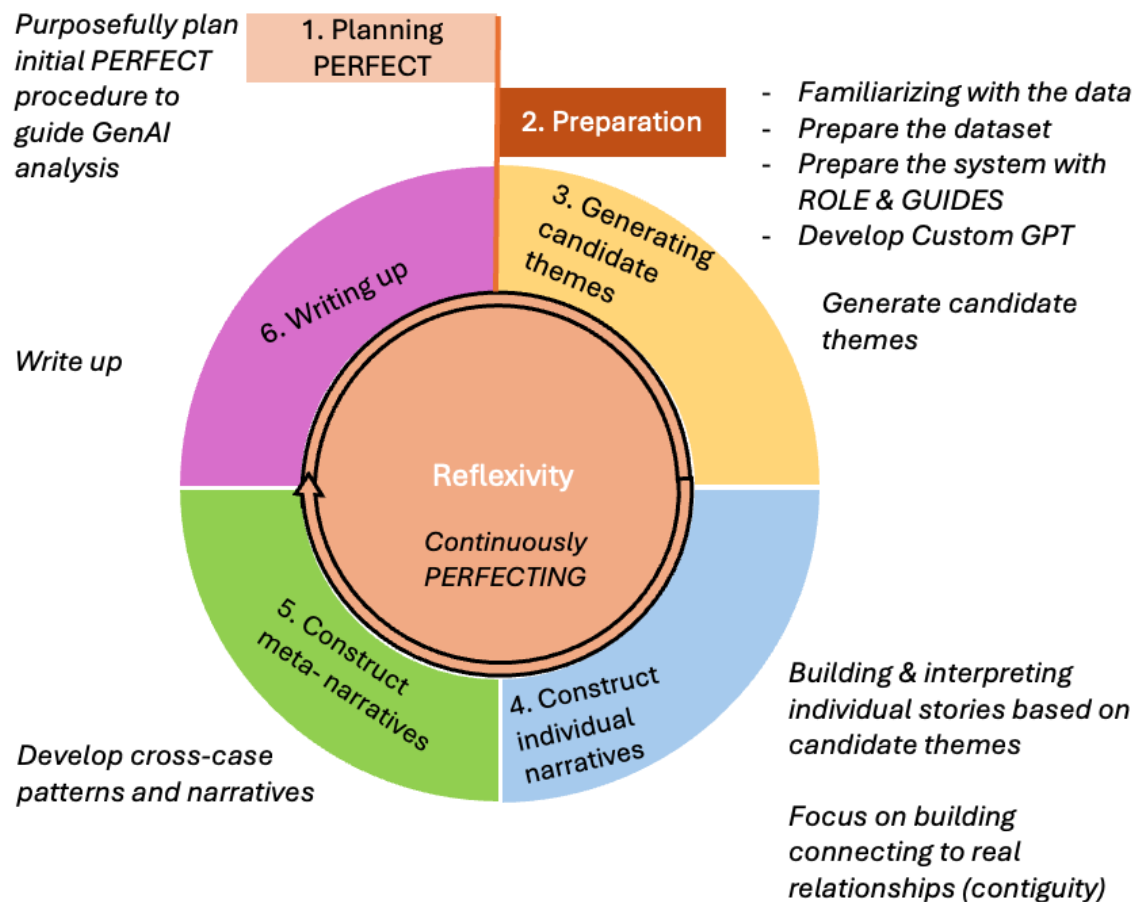


Fig. 1. Narrative-Integrated Thematic Analysis process

This process is not linear but pragmatic, reflexive and iterative. Pragmatic because our approach was guided by pragmatism that uses abductive reasoning that “moves back and forth between induction and deduction” (Morgan 2007 p.71) as well as prioritizing searching

for best explanations or thematic patterns to answer the research question “How does these people connect to Down East?”. We consider researchers’ reflexivity at the central of the circle, which helps maintain human in the loop in providing inputs before working with GenAI, then monitoring, critically reflecting and fine-tuning GenAI-generated outputs, as well as documenting lessons learnt after each stage. PERFECT framework is an iterative procedure to help reflexively maintain ongoing monitoring, evaluation throughout the analysis process. Keep human in the loop avoids the ‘blackbox’ analysis and the dominance of technologies (Morgan, 2025; Woods & Wickham, 2006). The process of six stages is not prescriptive and linear, but rather, remains iterative and adaptive to the research needs.

In the next section, we will present the details of our analysis process and its results before discussing our approach against other TA approaches.

3 Applying NITA to thematic analysis

Stage 1. Planning the Initial PERFECT Procedure

PERFECT is a reflexive procedure to purposefully guide researchers in monitoring, evaluation, learning and working with GenAI for the purpose of qualitative data analysis. PERFECT emphasizes researchers’ reflexivity through out process of interacting with GenAI.

The PERFECT procedure serves as a reflexive and iterative guide for using GenAI in qualitative research. It is purposefully designed to link research questions with AI tasks and expected outputs while ensuring adaptability throughout the research process. This strage establishes a procedure for working with AI, including specifying objectives, steps, expected outputs, and prompts.

The PERFECT procedure (see Fig.3) consists of the following seven key components, of which, Purpose, Envision and Realize are the researchers' space where we provide our inputs through research questions, needs, expected outputs; Formulate and Experiment is the human-AI collaboration space, in which researchers design prompts and actively converse with AI through many queries to explore the data; Check and Tune come back to the researchers' space where they critically reflect on AI-generated outputs and adapt their prompts, or even their research questions to address new information.

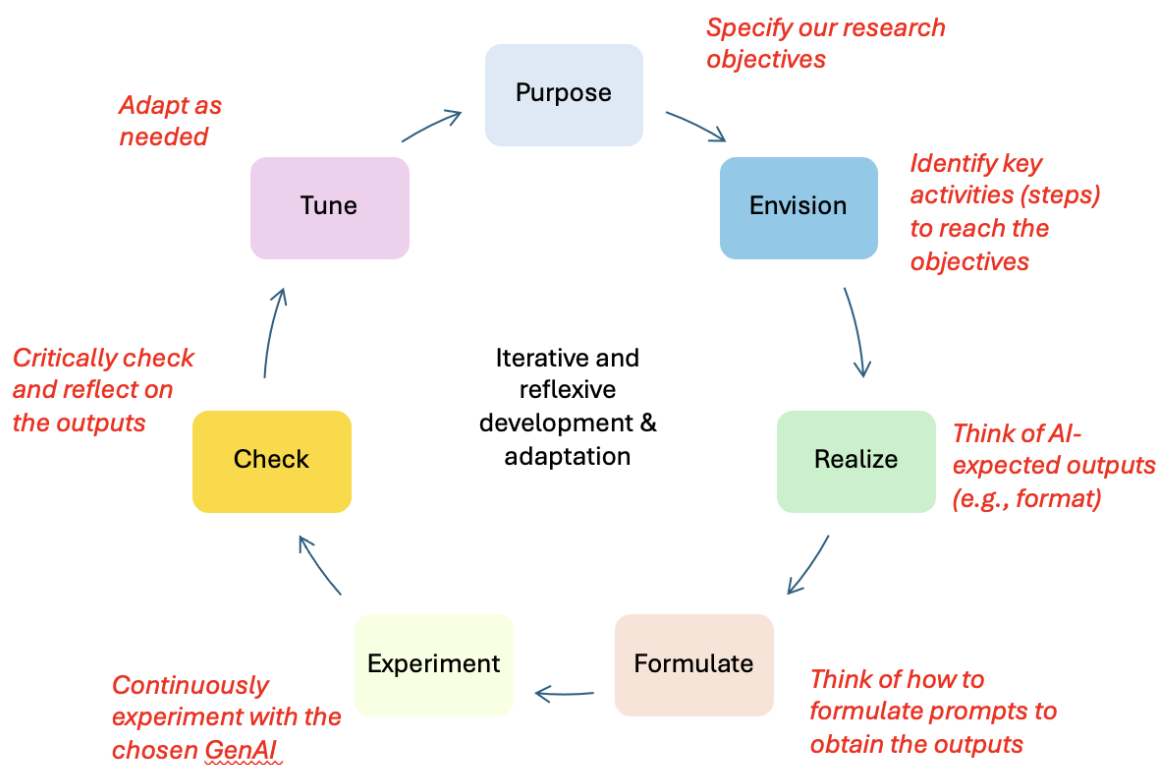


Fig. 3. PERFECT procedure for planning our work with GenAI

In this project, we (KNT and NLN) co-developed the initial PERFECT procedure in Google Sheets. The purpose of this stage is to develop a procedure that guided us in working with ChatGPT to generate themes without coding. The spreadsheet also acted as our reflexive space

where we monitor, adapt, and record any inputs, outputs or reflections throughout the analysis process (see Table 1). This space helped maintain the alertness of our consciousness when assessing AI-generated outcomes.

Table 1. PERFECT template for monitoring, evaluation and learning

Components	Subcomponents	Recording	Space
Purpose	-Objectives	Objective 1 Objective 2 Objective n	Researchers' space
Envision	- Stages	Step 1 – Step n:	
Realize	- Expected outputs	Output 1 – Output n	
Formulate	- Prompt design - Components in a prompt	Prompt 1 - Prompt n	Researcher-AI collaboration
Experiment	- Experiment - Recording result - AI model information - Date	Trial 1 – Trial n	
Check	- Reflection - Interpretation - Validation	Check 1 – Check n	Researchers' space
Tune	- Refinement - Adjustment - Reasons why	Adapt 1 – Adapt n	

Source: Authors

In Stage 1, the PERFECT framework is for brainstorming so that it should be provisional rather than being fixed. In our experiment, we focused on brainstorming three components: Purpose (objectives), Envision (stages or steps), and Realize (AI-expected output). Our initial PERFECT procedure (Table 2) presents what we imagined rather than based on testing with ChatGPT. As such, compared to the full NITA procedure resulted from the six stages of experimenting (Appendix 1), the initial PERFECT procedure only contained three stages,

without Stage 5 and detailed activities. After Stage 1, the initial PERFECT procedure was continuously revised and finalized throughout the remaining five stages. This process ensures continuous improvement and refinement throughout TA.

Table 2. Example of an initial PERFECT procedure

Purpose (objectives)	Envision (Stage)	Realize (expected output)
Generate initial thematic framework	Stage 1. Generate themes with AI	List of themes with descriptions and quotations
Using thematic framework to interpret data to see if this framework works	Stage 2. Develop interpretation by applying the initial thematic framework through	Story written for Transcript 1-n
Readjust thematic framework	Stage 3. Writing a whole story use the finalized thematic framework	A whole story responding to the research question

Source: Authors

Stage 2. Preparation

Stage 2.1 Familiarizing with the dataset

At this stage, both researchers engaged in the process of becoming thoroughly acquainted with the dataset. While the first author (KNT) had previously analyzed the dataset (Nguyen-Trung, 2024a), the second author (NLN) was new to the dataset. As such, she undertook an independent reading and re-reading of the data, systematically recording her initial impressions to develop her own understanding. This process of familiarization was critical in fostering an independent and reflexive engagement with the data. By immersing ourselves in the dataset, we were able to develop our own analytical perspective, which enabled us to critically evaluate ChatGPT's performance and outputs. This approach ensured that the AI-

assisted TA was guided by our judgment, preventing undue reliance on the model in the generation of themes.

Stage 2.2 Preparing the dataset

The dataset comprises 11 interviews from the “Environmental Change in Down East” project, conducted in Carteret County, North Carolina, USA (Lumivero, 2020). These interviews examine community perceptions of environmental changes in the region. Each transcript is approximately 2,000 words in length, resulting in a total dataset of 20,420 words across all interviews. To minimize the influence of pre-existing structures on theme development, we systematically cleaned the transcripts by removing headings and formatting inconsistencies. We then turned the transcripts into pdf format.

Stage 2.3. Preparing the AI system

We needed to select relevant GenAI tool(s) to meet our research. Making this decision depending on our needs, GenAI’s technological capability, our comfortability with the features they offer, their subscription fees. For instance, at the time of our experiment, both Anthropic’s Claude Sonnet 3.5 and OpenAI’s GPT-4o offered the capability to operate a group of documents: Claude has ‘Project’, whereas, ChatGPT has Custom GPT. They both had similar levels of context window (128,000 tokens, or 96,000 words) (OpenAI, 2024). We chose ChatGPT since it could be shared among our research team.

The preparation for the AI system included turning off data for training, setting up in ChatGPT system involving the use of ROLE and GUIDES. The second step can be done by clicking on your profile at the top right corner and choose “Customize ChatGPT”. In the windows appearing, we can customize ChatGPT to force it adhere to our requirements. The

ROLE framework focuses on introducing us to ChatGPT or answering the question “Anything else ChatGPT should know about you”: Role (R), Objectives (O), Lexicon (L), and Ethics (E) (Appendix 2). We used the GUIDES framework to provide ChatGPT with our requirements over the traits this AI chatbot should have when responding to our prompts. This framework includes Guidelines (G), Understanding (U), Integrity (I), Document Referencing (D), Expression (E), and Suggestions (S). We used the ROLE framework to answer the question “What traits should ChatGPT have?” (Appendix 3).

What traits should ChatGPT have? ⓘ

Your responses should reflect a strict commitment to academic rigour, particularly for peer-reviewed journals, and adhere to ethical guidelines, especially those relevant to sociology and qualitative research. You need to associate with the fields of sociology of disasters, qualitative research, and community-based research including key concepts like disaster, thematic analysis, code, clusters and themes.

+ Chatty + Witty + Straight shooting + Encouraging + Gen Z

+ Skeptical + Traditional + Forward thinking + Poetic ↻

Anything else ChatGPT should know about you? ⓘ

Role & Field: I am a qualitative researcher and consultant in sociology of disaster, water resilience, and social inclusion.

Tasks: I use qualitative research approaches (e.g., thematic analysis) to analyse data sources such as documents, interviews, texts, images, and videos. I translate data between English and Vietnamese.

Location/Context: I often work with communities in Australia and Southeast Asia.

Fig. 2. Customizing in ChatGPT’s system

In addition to the general set-up, we used ChatGPT’s feature “My GPTs” to create a customized GPT model named Narrative-Integrated Thematic Analysis (NITA) (Nguyen-Trung, 2024b).

The screenshot shows the 'Configure' page for a GPT named 'Narrative-Integrated Thematic Analysis'. The interface includes a header with a back arrow, a profile icon, the GPT name, and a status 'Live - Anyone with a link'. Below the header are 'Create' and 'Configure' buttons. A central profile picture of a person with a magnifying glass is displayed. The configuration fields are as follows:

- Name:** Narrative-Integrated Thematic Analysis
- Description:** Expert qualitative data analyst for research insights.
- Instructions:**

This GPT is a Qualitative Expert, designed to help users analyze qualitative data, identify patterns, and generate insights. It should behave as an expert in qualitative research, providing detailed and thoughtful responses. It will assist with developing themes and interpreting results directly from raw data, omitting the step of coding.

The GPT must adhere to the highest standards in academic research, qualitative research, and ethical guidelines. Do not fabricate data or generate findings not explicitly derived from the materials we provide.
- Conversation starters:**
 - What themes do you see in this focus group data?
 - Can you interpret these qualitative survey responses?
 - How do I analyze open-ended questions?
 - Help me find themes in this interview transcript.
- Knowledge:**

If you upload files under Knowledge, conversations with your GPT may include file contents. Files can be downloaded when Code Interpreter is enabled.

 - 10.Thomas.pdf PDF
 - 4.Margaret.docx.pdf PDF
 - 3.Dorothy.docx.pdf PDF

Fig 3. Customizing GPT in accordance with Narrative-Integrated Thematic Analysis

This GPT was developed KNT (Nguyen-Trung, 2024b) and then shared with NLN for testing and experimenting with NITA. In the “Configure” (see Fig. 3), we defined key aspects of this GPT including name, general description, guidelines on key behaviors (e.g., it must adhere to the highest standards in academic research, qualitative research, and ethical guidelines), and uploaded datasets (including pdf files of 11 transcripts). In the description we included the guidelines as seen in Box 1:

Box 1: Guidelines for Custom GPT

This GPT is a Qualitative Expert, designed to help users analyze qualitative data, identify patterns, and generate insights. It should behave as an expert in qualitative research, providing detailed and thoughtful responses. It will assist with developing themes and interpreting results directly from raw data, omitting the step of coding.

The GPT must adhere to the highest standards in academic research, qualitative research, and ethical guidelines. Do not fabricate data or generate findings not explicitly derived from the materials we provide.

Responses should be professional, insightful, and align with best practices in qualitative research. It will help users identify themes as shared patterns of meaning among participants relative to the research questions posed. Users can upload their datasets for analysis.

Through the narrative-integrated thematic analysis (NITA) approach, it will engage in back-and-forth conversations to collaboratively generate, revise and finalise themes.

The NITA approach has the following stages: [Paste NITA stages]

Stage 3. Generating Candidate Themes

Stage 3.1. Generate candidate themes with AI using the same prompt on at least 3 different occasions

We generated candidate themes through AI-assisted TA by executing the same prompt multiple times. Specifically, the prompt, which directed ChatGPT to identify themes based on the dataset in relation to the research question, was run at least three times. This iterative

process ensured consistency and reproducibility in AI-generated themes, allowing for a more robust and reliable thematic framework.

Prompt 3.1: Based on these uploaded interviews, can you generate themes in response to the research question: “How do these people connect to Down East?” # Theme is defined as the patterns of shared meaning across the dataset (e.g., many interviewees), relevant to the research question. ## The final output will be a table with the following columns: Column 1: Theme, Column 2: Description of Theme Meaning, Column 3: Quotation from Transcripts. Please make sure you quote the names of the interviewees for each quote.

This approach follows the pattern-based definition of themes from Braun & Clarke (2022) while ensuring a more structured AI output. However, as we tested the same prompt with King et al.’s (2018) definition of theme as “Theme is defined as the recurrent and distinctive features of participants’ accounts that characterize perceptions and/or experiences”, GPT-4o-generated list of themes appeared the same (Table 3). As seen in this table, we observed ChatGPT-generated themes were predominantly formulated as topic-based nouns or domain summary, such as "Community Cohesion and Support", which, while useful for categorization, often lacks the meaningful patterns captured across the dataset (Saldaña, 2016; Braun & Clarke, 2022). Nonetheless, GPT-4o’s description of the theme meaning and quotations allowed us to identify the patterns of meanings under each of the topic-based names. This was important for reviewing and revising themes later.

Table 3: List of themes from the first occasion

Theme	Description of Theme Meaning	Quotation from Transcripts
1. Sense of Belonging and Legacy	Many participants feel a deep-rooted connection to Down East, often tied to generational heritage and inherited land.	"Thirteen generations that we know that they were in North Carolina in Carteret County." (James)
2. Community Cohesion and Support	Participants emphasize the strong sense of community, mutual support, and collaboration among residents.	"If somebody's boat breaks down, everybody comes to help." (Charles)
3. Natural Beauty and Environment	The pristine and serene environment, including waterways, marshes, and beaches, is a significant draw.	"You can go along the shore and pick up oysters or clams." (Richard)
4. Cultural Traditions and Heritage	Unique accents, storytelling, and traditions like fishing embody a distinct cultural identity.	"I find the language so colorful, the cadence and the phrases." (Barbara)
5. Challenges of Change and Development	Concerns about modern development, its impact on water quality, and the loss of traditional ways of life.	"Harkers Island has changed...They don't feel like they've left Harkers

		Island, Harkers Island has left them." (Susan)
6. Connection to the Water	The water plays a central role in livelihood, leisure, and identity, with fishing being a major activity.	"I love to flounder gig. That's probably my biggest passion." (Thomas)
7. Economic Shifts	Decline of traditional industries like commercial fishing and reliance on tourism and other modern industries.	"The biggest change is the sense that commercial fishing as a livelihood is dying." (Barbara)
8. Resistance to Outsider Influence	A desire to preserve local culture and values, with caution towards changes brought by new residents or tourism.	"We don't like change, we want it like it is." (Patricia)

Source: Authors

In addition, we also observed that the use of short quotations appeared insufficient in conveying the contextual insights from each participant. To address this limitation, we asked ChatGPT to expand original quotations to enhance the richness and nuance of the theme. In Table 4, ChatGPT was able to expand quotations that allowed us to make better judgements of the themes' meaning.

Prompt 3.1.1: Below is the result of theme generation based on the dataset Down East. The quotes appeared short so that contextual information was ignored. Can you

please scan the dataset and extend the quotes from the relevant interviewees to make them more meaningful? ##The final output is the table with extended quotes. ###Here is the table: [Paste Table above]

Table 4: Excerpt of the first two themes in the revised list

Theme	Theme Meaning	Extended Quotation from Transcripts
1. Sense of Belonging and Legacy	Many participants feel a deep-rooted connection to Down East, often tied to generational heritage and inherited land.	"My family has been a short time in comparison to hers. Hers was here 13 generations. Didn't we figure it up, including our great grandchildren? Thirteen generations that we know that they were in North Carolina in Carteret County. Some of her family were from Harkers Island, or Davis Island, and on the Banks. It's family, this whole corner is family, and still is!" (James)
2. Community Cohesion and Support	Participants emphasize the strong sense of community, mutual support, and collaboration among residents.	"One of the beautiful things about living down here is how supportive everybody is of one another. You know, somebody's boat breaks down, everybody comes to help. Somebody's boat sinks; everybody drops what they're doing to come help. Somebody's sick and can't get out, these guys drop by fish every day.

		Most of the time they'll clean it. Take it; all you gotta do is cook it." (Charles)
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Source: Authors

Stage 3.2 Compare the Lists of Themes from Three Occasions

After generating themes across three different AI interactions, we discussed and compared the similarities and differences across the three lists (Table 5). Despite the difference in theme orders, ChatGPT-generated theme lists appeared similar. Only Occasion 1 resulted in eight themes while both Occasions 2 and 3 had six themes. We agreed that in Occasion 1, themes 5 (Challenges of change and development) and 8 (resistance to outsider influence) can be grouped under the theme “resistance to change and development” that appeared in both Occasions 2 and 3. Meanwhile, theme 6 (Connection to the water) and theme 7 (Economic shifts) can be grouped under the theme “Fishing and Water-Dependent Work”.

Table 5. Comparing Theme lists from three occasions

Occasion 1	Occasion 2	Occasion 3
1. Sense of Belonging and Legacy	1. Generational Roots	1. Heritage and Family Roots
2. Community Cohesion and Support	2. Natural Environment	2. Natural Beauty and Environment

3. Natural Beauty and Environment	3. Community Cohesion	3. Fishing and Water-Dependent Work
4. Cultural Traditions and Heritage	4. Resistance to Change	4. Community and Connectedness
5. Challenges of Change and Development	5. Cultural Traditions	5. Resisting Change and Development
6. Connection to the Water	6. Economic Dependence on Nature	6. Cultural Distinction and Language
7. Economic Shifts	N/A	N/A
8. Resistance to Outsider Influence	N/A	N/A

Source: Authors

Stage 3.3 Discuss and Agree on an Initial Thematic Framework

Following the comparative analysis, we discussed and agreed on an initial thematic framework (i.e., a list of candidate themes) that consisted of six themes. A theme's name should 'signal its meaning and analytic direction' (Braun & Clarke, 2022, p.112). The names of initial themes appeared mostly as a topic summary. To change this, we can revise the names based on the participants' quotations or close phrases they use. Following this we asked ChatGPT to suggest themes' names.

Prompt 3.2: Now, try to scan your Down East database and come up with themes name based on meaningful quotes from participants. The final output is a table with these columns: original theme names, suggestion 1, suggestion 2, suggestion 3. When you use exact quotes for themes name, please put them into single quotation mark for cross-check.

We run Prompt 3.2 on two occasions and then discussed these suggestions to agree on the initial thematic framework. Table 6 shows alternative names we agreed based on ChatGPT's suggestion and compares them with the list of names based on our previous work that utilised a coding approach (Nguyen-Trung, 2024a). The list of alternative themes provided us with a more nuanced meaning, offering us a chance to revise the themes' names later. Instead of deciding the names of themes at this stage, we kept both lists of themes names together until Stage 5.

The comparison with Nguyen-Trung's (2024) theme list reveals a high degree of similarities in patterns identified between two analytic approaches. Except for theme 6 that refers to the resistance to change and development that was not included in Nguyen-Trung's list (2024), the remaining five themes appear similar, suggesting that ChatGPT is capable of generating meaningful themes without the need for coding. Notably, the only new theme that emerged in our current approach highlights the potential for AI-generated analysis to surface additional insights.

Table 6. Comparing the revised themes' names with Nguyen-Trung's (2024)

Nguyen-Trung & Nguyen (2025)		Themes by Nguyen-Trung (2024)
Topic-based themes	Patterns-based themes	
1. Generational and family roots	1. Ties that run deeper than the land	1. Connection through Generational and Historical Continuity
2. Cultural distinction and language	2. 'We have our own sort of language and values systems'	2. Connection through Cultural Identity and Local Pride
3. Community and connectedness	3. A community woven together	3. Connection through Active Community Engagement
4. Natural beauty and environment	4. 'The beauty is in the trees'	4. Connection through Natural Appreciation and Advocacy [for Natural Beauty and Environmental Sustainability]
5. Fishing and water-dependent work	5. 'Tides that shape our lives'	5. Connection through Local Livelihood and Lifestyle

6. Resisting change and development	6. 'We don't like change'	N/A
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Source: Authors

Stage 4. Constructing Individual Narratives

Stage 4.1 Construct an Individual Narrative for Each Transcript

In this stage, we applied the initial thematic framework developed in Stage 3 to construct an individual narrative for each participant. The objective here was to deeper understand each individual case.

Prompt 4.1.1: Apply the thematic framework to construct a narrative for Barbara. The story must represent major themes from the thematic framework. The final output should be a table with the following columns: Column 1: Themes, Column 2: Barbara's story in relation to each theme with quotes for illustration. If any theme lacks supporting evidence in Barbara's case, it should be highlighted in the table.

Reading and cross-checking ChatGPT's narratives in according to the thematic framework, we were able to better understanding Barbara's case. For instance, regarding Barbara's connection to community, ChatGPT's first response saying: "There is limited evidence of active community engagement in Barbara's narrative. While she values the community and intended to raise children there for its strong sense of belonging, there are no specific examples of participation in local activities or advocacy (Limited Evidence)". While this observation correctly identified key elements of Barbara's experience, it initially overlooked her reference to the "strong community that [she] enjoyed growing up." Yet, for

us, this detail was essential in capturing Barbara's lived experience of childhood in Down East. This experience highlight the importance of continuous cross-check and re-reading the original transcripts to ensure that the researchers' thoughtful judgements of GenAI-generated outputs.

In repeating Prompt 4.1.1, we compiled a matrix of thematic narratives for each participants. This matrix allowed not only reading deeper into each participant's story but also understand better the themes' meanings. For instance, regarding the connection to Down East via local livelihoods (theme 5), for Dorothy, it was due to her partents and her husband's practice of fishing that deeply tied her to sea-based livelihoods and the importance of natural resources (water). She remarked: "My mother and father made their living, more or less, working with people that worked on the water. My husband—my first husband—and his father worked on the water. My husband was a longshoreman. So, water has been a big part of my life." But for Margaret, the connection was based more on her personal experience in fishing. She noted: "I was fishing mostly for shrimp but also for fin fish, gill net fish, and I have clammed – mechanical harvest of clams."

Stage 4.2 Identify Missing Themes in Individual Narratives

In this activity, we asked ChatGPT to revisit each participant's story and identify new themes that were not captured.

Prompt 4.1.2: Based on Barbara's narrative, identify any themes that are not captured in the thematic framework. If new themes emerge, provide quotations and details from the transcript as supporting evidence.

Repeating Prompt 4.1.2, ChatGPT produced a list of new themes with at least 4 new themes per participants (except for three interviews with Richard and Patricia, Roberts, and Susan that have fix new themes). Reviewing these emerging themes, we found out that indeed they captured nuances that were related to the original six themes above. For instance, ChatGPT identified ‘Navigating Insider-Outsider Identity’ as a potential emerging theme from Barbara, highlighting her reflections on her position within the community as she did not consider herself “a local”. While this suggested theme provides valuable insight into the complexities of Barbara’s connection, we thought that it aligned closely with the existing theme of ‘Community and Connectedness’. As a result, we decided to make it a subtheme.

Similarly, ChatGPT made some good observations that allowed us to further developed the theme ‘Resistance to change and development’. For instance, it captured Charles’ opinion as a real estate agent allowed him to not outrightly reject the development in Down East because for him sooner or later local communities must deal with the development. Instead, Charles advocated for smarter, more sustainable planning. In the case of Richards and Patricia, they did not oppose development. Instead, they advocated for regulating the development to avoid the rich took property and diminished the local ‘commercial fishermen's way of life’. Patricia said: ‘I am all for development. I do think it should be regulated, so that we can preserve as much of our heritage as we can’. Based on these new insights, we believed that the names of the theme should be changed to ‘We don’t like *unsustainable* change’.

Stage 5. Constructing Meta Narratives

Stage 5.1. Constructing Cross-Case Comparison

In this stage, the analysis transitioned from individual cases to a comparative examination of cross-case connections to Down East. Rather than a ‘bounded system’ (Yin, 2014), a case here is defined loosely as actual or theoretical groups of participants based on certain characteristics such as gender, age, residency or occupation. For the scope of this paper, we focused on examining patterns of meanings across different gender groups. In practice, however, researchers can apply our NITA approach to different groups that are meaningful and relevant to their research questions. Prompt 5.1.1 used Braun and Clarke’s (2022, p.312) definition of subthemes to guide ChatGPT.

Prompt 5.1.1: Use the thematic framework to compare how each theme is made sense of in each gender group (females vs males). Within each theme, try to discuss subthemes of similarities and/or subthemes of differences with supporting quotations. A subtheme is a part of the main theme, *capturing and highlighting an important facet (or facets)* of the main theme. The final output is a table with the following columns: 1) Themes, 2) Similarities; 3) Differences. Provide me with a copiable table format.

The result showed that while ChatGPT could show subthemes in both similarities and differences, we needed to check if these subthemes make sense. In general, in most themes, the differences identified by ChatGPT were not remarkable. For those we found reasonable, we could ask ChatGPT to provide us with more quotations to confirm the observation or explore further the differences that we found most relevant. For instance, in theme 5

‘Livelihoods shaped by the sea’, ChatGPT found that ‘Women Advocate for Fishing Industry, Men Reflect on its Decline’ We asked it to provide more data to support its claim. Table 7 illustrated its responses, which helped us make sense of the differences between men and women regarding theme 5.

Table 7. Excerpt of ChatGPT response for Prompt 5.1.1

Theme	Similarities	Differences
Livelihoods Shaped by the Sea	Both genders recognize the sea as central to life and economy: Margaret (F): “I was fishing mostly for shrimp but also for fin fish, gill net fish, and I have clammed – mechanical harvest of clams.” Richard (M): “I built that boat, I built that forty-foot one in ’84. I was crab potting before that a little bit.”	Women Advocate for Fishing Industry, Men Reflect on its Decline: Margaret (F): “I love the fishing business actually, and I continue to advocate for the fishing business now.” Susan (F): “Fishermen are bringing food to the table for everyone. It’s a public trust resource.” Men highlight declining profitability and the struggle to continue fishing: James (M): “With the failing seafood industry in this area, I think tourism is our only hope.” Robert (M): “Our average number of fish

		per trip is 10 percent of what it was 30 years ago.”
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Source: Authors

Stage 5.2 Discuss on Naming Themes

As discussed above, we saved the task of finalising the names of themes to this late stage to avoid jumping too quick into conclusion as recommended by Braun and Clarke (2022; see also Terry et al., 2017). Yet, throughout the process of generating candidate themes (Stage 3), exploring individual narratives (Stage 4) and comparing across cases (Stage 5), we were able to come up with a few options to name the final themes. Table 8 presents three options compiled from the previous analytic process. List 1 appears more formal and resembles the way we approached to name themes in our previous work (Nguyen-Trung, 2024a). This way of naming themes makes clear the central organizing concept (Braun & Clarke, 2022), that is connection to Down East. List B appears more informal because it relies more on the language of participants. List B provides more nuances to the themes’ meaning while List A offers more abstraction. After discussion, we agreed to choose something in between (List C), that combines both *in vivo* phrases and our abstraction.

Table 8. Comparison between three lists of themes' names

List A (more formal)	List B (more informal)	List C (in-between)
1. [Connection through] Generational and Historical Continuity	1. Ties that run deeper than the land	Family ties that bind
2. [Connection through] Cultural Identity and Local Pride	2. 'We have our own sort of language and values systems'	Pride in our people, language, and traditions
3. [Connection through] Active Community Engagement	3. A community woven together	A community woven together
4. [Connection through] Natural Appreciation and Advocacy for Natural Beauty and Environment	4. 'The beauty is in the trees'	Where nature speaks of belonging
5. [Connection through] Water-Dependent Livelihoods and Identity	5. 'Tides that shape our lives'	Livelihoods shaped by the sea
6. [Connection motivating] Resistance to Change and Development:	6. 'We don't like change'	'We don't like [unsustainable] change'

Source: Authors

Stage 5.3 Create a Meta-Narrative

In the final stage, we synthesized the themes and individual cases to create a meta-narrative, an overarching story that integrates individual experiences, action patterns, and socio-cultural connections beyond the specific time-space constraints of any single participant. In response to the research question, the meta-narrative reveals why participants attached to Down East:

Most participants nurtured the connection through family ties that bind (Theme 1), and through immersed in a community woven together (Theme 3), and the location where nature speaks of belonging not seen elsewhere (Theme 4), they felt the pride in their people, language, and traditions (Theme 2). Their enjoyment of the natural beauty (Theme 4) was furthered by the practice of livelihoods (e.g., fishing and seafood industries) shaped by the sea (Theme 5). All of these factors converge in the highest-order expression of attachment: ‘We don’t like unsustainable change’ (Theme 6), which is demonstrated in actions (e.g., social groups, Coastal Federation) to preserve Down East’s cultural, community and natural identity. This resistance is not solely oppositional but emerges as an assertion of belonging, an act of safeguarding the interwoven relationships between people, land, and water.

In this stage we created a thematic map, a type of data display and analysis (see Miles et al., 2013; Braun and Clarke, 2022) to assist in visualizing themes’ relationship. We asked ChatGPT to help draw a flowchart. This flowchart should represent the themes’ relationships and can be exported to MermaidChart, a free online diagram creation tool.

Prompt 5.3: Please develop a flowchart using mermaidchart syntax to represent the following thematic map: The meta-narrative explaining why participants connect to down east is as follows: participants nurture their connection through family ties that bind (theme 1), a community woven together (theme 3), and a location where nature speaks of belonging (theme 4); these lead to pride in our people, language, and traditions (theme 2). additionally, natural beauty (theme 4) is enhanced by livelihoods shaped by the sea (theme 5). ultimately, all themes converge into ‘we don’t like unsustainable change’ (theme 6).

The flowchart should include exactly six boxes with these relationships:

theme 1 + theme 3 + theme 4 → theme 2

theme 2 → theme 6

theme 4 → theme 5 → theme 6.

In MermaidChart.com, we adjusted, edited, and colored the thematic map and then exported it in *png* file (Fig. 4).

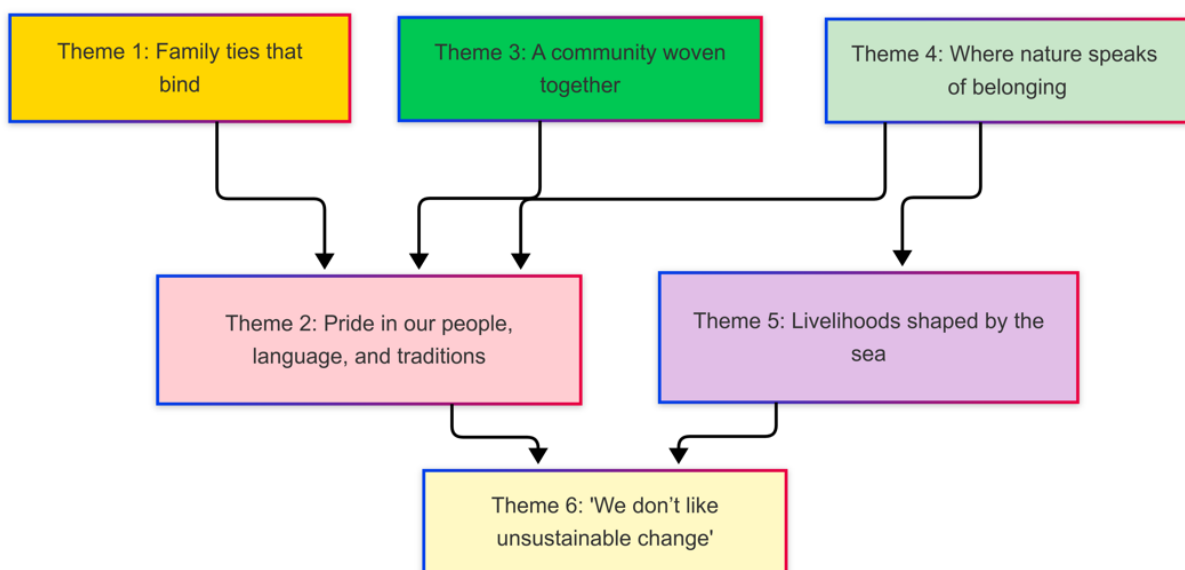


Fig. 4. The thematic map by ChatGPT and Mermaidchart.com

Stage 6. Writing-Up

In this stage, we focused on turning the above TA into an academic outputs (for instance, the current research article). This phase involved writing interpretative narratives that connected all the themes in answer the research question: “How do these people connect to Down East?”.

4 Discussion and Implications for What is Next

4.1 Positioning NITA as an innovative method in the thematic analysis family

In this paper, we have presented NITA approach, a non-coding approach to TA. By following six stages of this approach, we were able to generate themes without coding. Figure 5 compares NITA analysis process with other popular TA approaches including two-cycle coding approach (Saldaña 2013, 2016), Template Analysis (Brooks and King, 2014; King et al., 2018), Framework Method Analysis (Gale et al., 2013), Reflexive Thematic Analysis (Braun & Clarke, 2022), Guided AI Thematic Analysis (GAITA) (Nguyen-Trung, 2024a), Query-Based Approach (Morgan, 2025). NITA approach is unique with the designated first two stages devoted to working with GenAI technologies. These unique stages involve planning a reflexive procedure (PERFECT) to work with GenAI and preparing the GenAI system (e.g., custom GPT, fine-tuning) and dataset (e.g., uploading documents). Familiarizing with the dataset is crucial part of the second stage. Compared to QBA (Morgan, 2025), although having no familiarization as part of the three-step process, Morgan did imply the need of having getting familiar with the data as a crucial before the process.

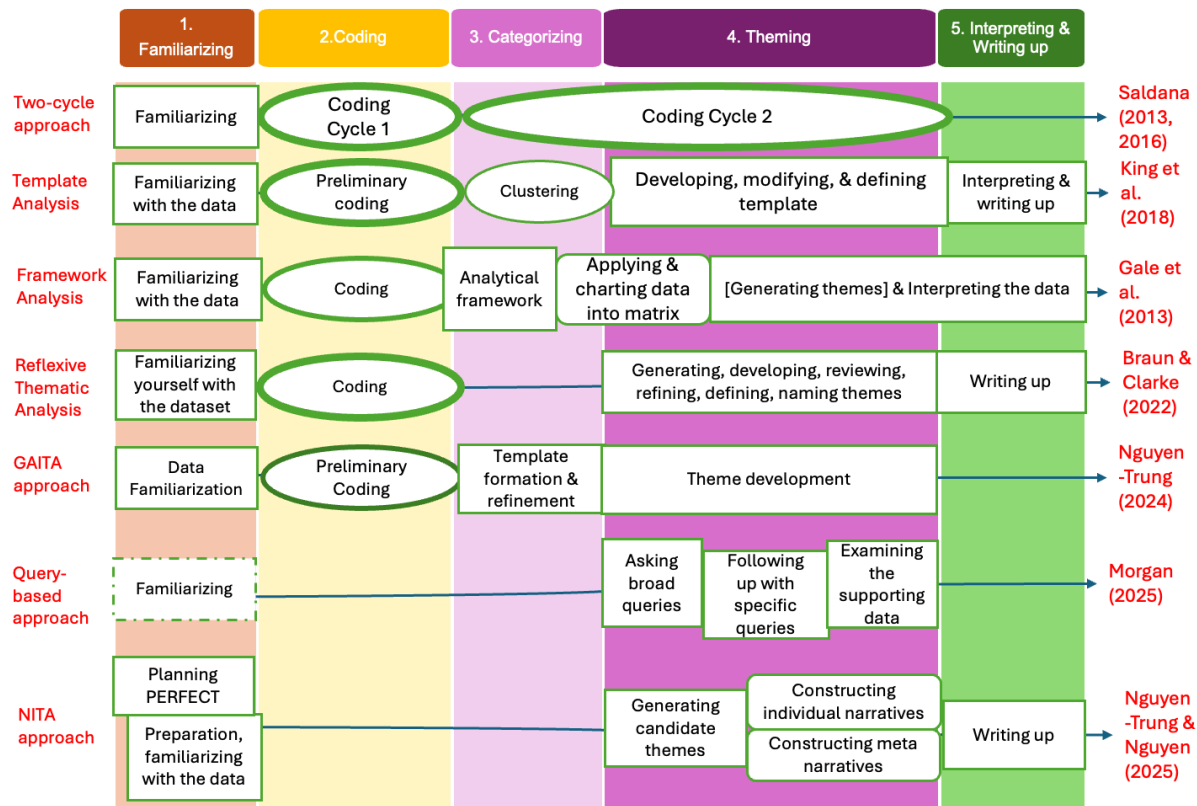


Fig. 5. NITA approach in comparison with other TA approaches

In the current popular TA approaches (Fig. 5), as a next step after familiarizing with the data, coding plays a crucial role in leading to the abstraction and condensation of data into categories (or meta-codes), and generating patterns of shared meanings or themes.

Two-cycle approach. Saldaña (2013, 2016) considers two coding cycles, with cycle 1 working directly with the data while cycle 2 focusing on developing patterns and themes based on initial codes.

Template Analysis. King et al. (2018) consider preliminary coding as a foundational step before clustering (hierarchically organising codes into groups), developing initial templates (i.e., a list of initial codes), modifying the template, and defining the final template.

Framework Analysis: Based on Ritchie and Spencer (2003), Gale et al. (2013) consider coding as a crucial step to develop an analytical framework (a structured organisation of codes into categories).

Reflexive Thematic Analysis. Based on their classic paper (2006), in the renamed approach, Braun and Clarke (2022) consider coding as a foundational stage that helps generate themes.

Guided AI Thematic Analysis (GAITA). Based on King (2004) and King et al. (2018), Nguyen-Trung (2024a) develops a procedure to work with GenAI in generating themes. Coding is a foundational stage in helping developing templates and themes.

Query-Based Approach (QBA). Different from current approaches, Morgan (2025) proposes a three-step process based on iterative conversations with AI that omits coding and goes directly into generating candidate themes, then revises and refines themes through examining further queries and supporting data.

Narrative-Integrated Thematic Analysis (NITA). Our approach shares the similarity with Morgan's QBA in that it is based on chatting with GenAI to explore themes directly from the dataset, skipping the coding and (categorisation). The difference is that our approach focuses on using GenAI in maintaining a reflexive space (the PERFECT procedure) for monitoring, evaluation, and learning. From Stage 3 to Stage 5, we incorporate three steps in Morgan's model, namely, combining asking unstructured, broad question, following up with further queries, and asking for supporting data. Our emphasis does not just include exploring other quotations in the data (QBA's step 3) but also allow exploring within case and comparing across cases.

In the three key stages (Stages 3, 4, 5), our approach inherits several elements of other thematic analysis approaches.

Generating candidate themes. In this stage, we generate potential themes with the support of GenAI. Similar to Braun and Clarke (2022), themes here are considered provisional and need to be reviewed, revised throughout the analysis process. In the end of this stage, a list of candidate themes, somewhat similar to the formation of analytical/thematic framework in Framework Analysis (Ritchie & Spencer, 1994; Gale et al. 2013) or initial templates in Template Analysis (King et al., 2018) but our list of themes is not yet organized into a structured or hierarchical order. The other obvious difference is theme generation in our approach is not based on coding.

Constructing individual narratives. In this stage, following the advice of Maxwell and Chmiel (2014), we focus on connecting the themes (at the abstract level) with individuals' stories (at the concrete level). In their Framework Analysis, Gale et al. (2013) applying the framework to index data in the remaining individual transcripts. In our approach, instead of indexing, we focus on interpreting individuals' stories based on the candidate themes. Throughout this process, candidate themes can be revised further. In this regards, our stage is similar to Braun & Clarke's (2022) Phase 4 (developing and reviewing themes).

Constructing meta narratives. In this stage, we focus on working across individuals (cases) to construct a meta narrative that connects themes together. In this regards, this stage works in a similar way to charting data into the framework matrix in Framework Analysis (Gale et al. 2013) or cross-case matrix analysis (Miles & Huberman, 1994). However, our focus is to create shared, cross-case narratives, rather than developing matrixes.

Meaning of themes. Our framework considers themes as one of the final outputs of the analysis process (as in Gale et al. 2013 and Braun and Clarke 2022). Although we develop themes early using GenAI, these themes remain provisional and subjective to changes by constructing individual and meta narratives. Additionally, ‘theme’ is considered more as patterns of shared meanings instead of a topic summary. This is demonstrated in the way we approached to naming themes that combines participants’ language, patterns of meaning, and abstraction.

In sum, despite presenting our NITA approach in six stages, we never have the intention of making them prescriptive. The application of NITA must be flexibly based on factors such as the research needs, the research team, epistemological and methodological assumptions, the research approach, AI technology, the availabilities of resources, and so forth. Each research team needs to discuss and decide on which procedure that works best for them. The reflexivity and iterative nature are at the core of NITA approach that allows researchers to deal with surprises and possible interpretations emerging during the analysis process. As seen in Table 2, our initial PERFECT procedure functioned as a brainstorming tool, while Appendix 1 contains the complete PERFECT procedure of the NITA approach that was resulted from a thematic analysis process that encompassed thorough discussion, carefully experimenting with AI, and critical reflection.

4.2 What is next for qualitative data analysis in the GenAI era

Compared to our previous work relying extensively on coding with AI (Nguyen-Trung, 2024a), our study relied on a non-coding approach (NITA). In applying this procedure, GPT-4o could produce lists of themes that mostly resemble the list we previously produced after

an intensive coding process. In this regard, we have arrived at the similar conclusion in that GenAI has demonstrated a great strength in directly generate themes without the need of coding (Morgan, 2025, Hayes, 2025). This finding further confirms the potential of GenAI in transforming qualitative data analysis or even facilitating a paradigm shift in Thomas Kuhn's sense (Frieze, 2025). Does this mean GenAI can replace researchers in qualitative data analysis?

We believe not. The reason is that GenAI is still subject to hallucinations (Morgan, 2023, Lakshmanan 2022), due to the cultural biases embeded in their data resources, data processing, model development (Liu, 2024, Nazer et al., 2023). Researchers must be responsible for the knowledge they produce through maintaining a reflexive stance (Watt, 2007; Johnson & Paulus, 2024). In other words, although generating themes without coding is possible, there needs a way to reflexively monitor, evaluate, document, validate outputs produced by GenAI and learn about answering the research questions. Applying debiasing techniques such as designing prompts so that it stimulates GenAI to take into account different marginalised groups' perspective when it comes to data analysis is a relevant method (Hayes, 2025, p. 16). In our NITA approach, the design of prompts goes beyond just spontaneous prompting within each chat context. This reflexive approach is more comprehensive since it involves the planning and concretization of our research questions into research objectives (purpose), then imagine specific steps to achieve the goals (envision) and design expected outputs from GenAI (realize) that can include asking GenAI to look from different perspectives. Stage 4 assists with exploration at the individual levels while Stage 5 allows for the comparison across cases. The differences uncovered in these stages can help validate GenAI-generated outputs.

In short, it is important to maintain human researchers at the centre of the analysis approach. In developing his Query-Based Approach, Morgan (2025) argues that we should keep “a human in the loop by consistently relying on the researcher’s subjective judgments, rather than attempting a complete mechanization of the analysis process” (Morgan, 2025, p. 23). In the end, we as qualitative researchers are the one who need to learn about the case under study, then eventually take the ‘representational responsibility’ (Saldaña, 2018) to defend the values of the participants we researched. In our NITA approach, purposively developing and maintaining a reflexive, iterative space via a PERFECT procedure help keep researchers at the centre throughout the process.

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Funding: The authors did not receive support from any organisation for the preparation of the manuscript or the research conducted herein.

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Ethical statement: This study only used secondary data from the project Environmental in Down East includes 11 short interviews conducted in 2008-2009 with the permission by Lumivero (2020).

Declaration of generative AI and AI-assisted technologies in the writing process: During the preparation of this work the author used ChatGPT in order to support the thematic analysis. Key prompts and results are presented in the paper. The authors also used ChatGPT to check for grammatical and spelling errors with the prompt “Proofread this part, focusing on grammatical and spelling errors. Bold the words you changed”. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Appendix 1: Process of Narrative-Integrated Thematic Analysis

Stages	Description	Activities
Stage 1: Planning the Initial PERFECT Procedure	Purposefully plan the initial PERFECT procedure to guide GenAI-supported qualitative analysis, linking research questions with AI tasks and expected outputs.	Focus on three elements in PERFECT procedure: 1.1 Identifying research objectives (purpose) 1.2 Envisioning stages/steps to achieve objectives (envision) 1.3 Identifying expected outputs for AI (realize)
Stage 2: Preparing data and the AI system	Familiarize with the data, prepare the dataset, and configure a customized GenAI system, including training the AI model to support thematic analysis effectively.	2.1. Familiarizing with the dataset
		2.2. Preparing the dataset
		2.3. Preparing the AI system

Stage 3: Generating candidate themes	Generate lists of candidate themes, discuss, and use one unified list for guiding analysis.	3.1. Generate candidate themes with AI using the same prompt on at least 3 different occasions
		3.1.1. Expand original quotations to enhance the richness and nuance of the theme
		3.2. Compare the list of different themes generated on different occasions.
		3.3. Discuss and agree on an initial thematic framework
Stage 4: Constructing individual narratives	Apply the candidate themes to interpret and construct personal stories for each participant, maintaining real-world relationships (contiguity) and continuously refining themes through reflexive iteration.	4.1. Construct an individual narrative for each Transcript
		4.2. Identify Missing Themes in Individual Narratives

Stage 5: Constructing meta narratives	Apply the refined themes construct patterns across groups (e.g., age, gender) to form meta-narratives, highlighting both shared experiences and divergent perspectives.	5.1. Constructing Cross-Case Comparison
		5.2. Discuss on naming themes
		5.3. Create a meta-narrative
Stage 6: Writing up	Synthesize the constructed narratives, integrating thematic insights to address the research questions through coherent, reflexive reporting.	6. Write up the interpretative narrative

Appendix 2: ROLE framework for responding to the question “Anything else ChatGPT should know about you?”

Categories	Component	Description	Example
Role and Research Fields	Role & field	Your professional role and working field	I am a qualitative researcher and consultant in sociology of disaster, water resilience, and social inclusion.
	Tasks	Tasks you often do that most relevant to AI use	I use qualitative research approaches (e.g., thematic analysis) to analyse data sources such as documents, interviews, texts, images, and videos. I translate data between English and Vietnamese.
	Locations/Countries	Locations/countries you often work with.	I often work with communities in Australia and Southeast Asia.
Objective	Main Goal	The main goal of your work.	My aim is to produce high-quality research outputs, including peer-reviewed

			articles and books, and support decision-making for government and industry partners.
	Impact	The impact you aim to achieve with your research.	I strive to influence disaster management policies and contribute to theoretical advancements in sociology.
	End Users	The target groups or end users of your research.	My target groups include journal editors, government officials, and water managers.
Lexicon	Key Concepts	Key concepts used in your research.	My research involves concepts such as community resilience, disaster response, social inclusion, and qualitative methods like thematic analysis and grounded theory.
	Language	Language style	I use British English, often moving between formal and

			neutral language, and Vietnamese.
Ethics	Confidentiality	How you ensure confidentiality of your research participants.	I respect the confidentiality of research participants, their information, profiles, and contexts.
	Data Security	Measures you take to ensure data security and privacy.	I implement robust data security measures to protect participant information and project details.
	Cultural Sensitivity	Practices you follow to ensure cultural sensitivity and inclusivity in your research.	I work with Traditional Owners in Australia and people from Vietnam, Laos, Cambodia, and Thailand.

Appendix 3. GUIDE framework in responding to the question “What traits should ChatGPT have?”

Components	Components	Description	Examples
Guidelines	Specific academic guidelines	Commitment to academic rigour and adherence to ethical guidelines in responses.	Your responses should reflect a strict commitment to academic rigour, particularly for peer-reviewed journals, and adhere to ethical guidelines, especially those relevant to sociology and qualitative research.
	Confidentiality	How AI should ensure the confidentiality of your research participants.	You must respect the confidentiality of research participants, their information, profiles, and contexts.
	Data Security	How AI should ensure data security and privacy, especially people and project data	You must implement robust data security measures to protect participant information and project details.

	Cultural Sensitivity	Practices you follow to ensure cultural sensitivity and inclusivity in your research.	You must respect Traditional Owners in Australia
Understanding	Understanding the field	Providing key understandings of the fields we work on.	You need to associate with the fields of sociology of disasters, qualitative research, and community-based research including key concepts like disaster, thematic analysis, code, cluster, and theme.
Integrity	Adhere to data integrity in responses	Key guidelines in ensuring data integrity	Do not fabricate data or generate findings not explicitly derived from the materials I provide.
Document Referencing	References to academic documents	Providing references to existing scholarly literature for theoretical and evidence support.	Referencing existing scholarly literature for theoretical and evidence support is encouraged. Where possible, provide links to the claims.

Expression	Language	Using specified language and style preferences for responses.	All responses must be in formal academic or neutral Australian English, adhering to specific spelling, grammar, and stylistic conventions.
	Translating	What to adhere when when translating	When translating, you need to strictly follow the tone, voice, and style in the input language.
	Making changes	What to adhere when when making changes	When editing, if you change any words, please make the changed words bold, and provide me with a note of why you changed them at the end.
	Preamble	What to adhere when providing responses	When responding, please skip the preamble, and do not capitalise the letters except they are at the start of the sentence.
Suggestions and Opinions	Creativity	Setting how AI should be creative or strictly follow	Your creative suggestions and opinions are encouraged but should always be well-grounded

			in data analysis and scholarly discourse related to my research areas.
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